

VARDHAN BRAHMANAPALLY

+1 804-298-0641 | vardhanchinnu52@gmail.com | [LinkedIn](#) | Richmond, VA

SUMMARY

Data Engineer with 3.5 years architecting and optimizing large-scale cloud data pipelines on Azure, with expertise in distributed computing, ETL/ELT development, and data platform modernization. M.S. in Computer Science (GPA: 4.0) from Virginia Commonwealth University (Jan 2025 – May 2026), with hands-on experience in ML engineering and agentic AI systems.

TECHNICAL SKILLS

Languages: Python, PySpark, SQL, Java, JavaScript, C++

Data Engineering: ETL/ELT Pipeline Development, Distributed Computing, Data Modeling, Dimensional Modeling, Data Profiling, Data Quality & Governance

Cloud & Platforms: Azure (Databricks, Data Factory, Data Lake, SQL DW, Logic Apps, DevOps), Microsoft Fabric, Snowflake, Delta Lake

Big Data & Processing: Apache Spark, PySpark, Query Optimization, Shuffle Reduction, Partitioning

ML & AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangGraph, LangChain, OpenAI API, NumPy, Pandas

DevOps & Tools: Azure DevOps, CI/CD Pipelines, SonarQube, Git, Power BI, Tableau

EXPERIENCE

Senior Data Engineer | Client: Procter & Gamble

Hyderabad, India

LTIMindtree ★ Shooting Star & Super Crew Award

Aug 2021 – Dec 2024

- Architected and led end-to-end migration of complex on-premises ETL workflows to a fully cloud-native stack on Azure Databricks and Azure Data Factory, standardizing data ingestion across disparate sources and eliminating legacy infrastructure bottlenecks.
- Designed a metadata-driven ETL framework enabling reusable ingestion pipelines across multiple regional business domains, reducing new pipeline development effort by over 60%.
- Architected Bronze-Silver-Gold Delta Lake pipelines supporting scalable, governed analytics while maintaining schema evolution, ACID compliance, and high-performance querying.
- Re-engineered regional data hub and mart architecture, delivering a 90% reduction in ETL processing time and supporting an 8x increase in data volume without degradation in pipeline performance or SLA compliance.
- Optimized Spark jobs through advanced query tuning, predicate pushdown, and shuffle reduction, achieving an additional 57% reduction in processing time across production pipelines.
- Implemented CI/CD pipelines using Azure DevOps and SonarQube, enforcing automated code quality gates and streamlining deployment cycles for a team of 4 engineers.
- Led data profiling and source-to-target mapping documentation for multi-regional pipelines, establishing reusable patterns adopted across the team.
- Built Power BI dashboards tracking ETL pipeline SLAs, data quality KPIs, and processing utilization across regional hubs, providing real-time observability to business stakeholders.

Software Engineering Intern

Hyderabad, India

Optival Solutions

Mar 2021 – Jul 2021

- Developed front-end features using React and JavaScript, and implemented backend API connectors using Java Servlets, supporting end-to-end feature delivery for the company's web platform.

PROJECTS

StockSense – Multi-Agent Financial Data Platform

Python, FastAPI, Pandas, Multi-Threading

- Built a 6-agent pipeline (Data, Technical, Fundamental, Sentiment, Decision, Risk), integrating quantitative signal generation with FinBERT-based sentiment scoring and LLM-driven decision synthesis into a unified trading signal.
- Engineered a historical backtesting engine (pandas, ta) to compute performance metrics (Sharpe ratio, max drawdown, win-rate) without lookahead bias.
- Optimized backend data throughput via parallel threading for high-frequency API data ingestion, cutting data-fetch latency by 60%.

Weather Along Route – Geospatial Data Pipeline

React, Mapbox GL, Open-Meteo API

- Engineered a time-dependent data pipeline that samples route coordinates, projects ETAs, and fetches time-matched hourly weather data across multiple third-party APIs.
- Designed a parallelized grid-based data aggregation system, partitioning route bounding boxes into geospatial cells and batching API queries to avoid rate-limiting while rendering a real-time heatmap overlay.

EDUCATION

Master of Science, Computer Science — GPA: 4.0

Jan 2025 – May 2026

Virginia Commonwealth University, Richmond, VA

B.Tech, Computer Science Engineering — CGPA: 7.7

2017 – 2021

Mahatma Gandhi Institute of Technology, Hyderabad, India

CERTIFICATIONS

Microsoft Certified: Data Engineer Associate — Score: 87.6% | Databricks Certified Associate Data Engineer — Score: 93%

Data Science for Engineers — NPTEL IIT Madras (Silver Medal, 81%) | Joy of Computing using Python — NPTEL IIT Madras (Gold Medal, 96%)

Introduction to Artificial Intelligence — NPTEL IIT Madras | SQL — Oracle | Google Cloud Computing Foundations — NPTEL IIT Kharagpur